AI-Enabled Smart Surveillance System using IoT

Anupam Kumari

MCA Student

K.R. Mangalam University, Gurugram Sohna, India

anupam8091@gmail.com

Abstract —This paper presents an AI-enabled smart surveillance system that integrates Internet of Things (IoT) and edge computing for real-time monitoring. Traditional surveillance systems mainly record video and require continuous human monitoring, which is inefficient and may delay the detection of security threats.

The proposed system uses OpenCV for face detection and DeepFace for face recognition to identify known and unknown individuals. The system processes video locally using edge computing, which reduces latency and improves performance. When an unknown person is detected, the system captures the image, stores it, and sends alerts to the user through email and Telegram notification.This system enhances security by enabling automated monitoring, faster decision-making, and reduced human intervention. It provides an efficient and intelligent surveillance solution suitable for modern security applications.

Keywords -AI, IoT, Smart Surveillance, Face Detection, Face Recognition, OpenCV, DeepFace, Edge Computing.

I. INTRODUCTION

The rapid increase in security threats, unauthorized access, theft, and criminal activities has made surveillance systems an essential part of modern society. Residential areas, offices, and public infrastructures require continuous monitoring to ensure safety and prevent potential risks. Traditional surveillance systems mainly rely on continuous video recording and manual monitoring, which is time-consuming and inefficient. These systems often fail to provide real-time responses, making them less effective in critical situations [1].

The limitations of manual monitoring highlight the need for intelligent and automated surveillance systems. Human operators may overlook important events due to fatigue or distraction, especially when monitoring multiple video feeds simultaneously. This reduces the reliability of traditional systems and increases the risk of security breaches. Therefore, there is a growing demand for systems that can automatically detect and respond to suspicious activities [2].

With advancements in Artificial Intelligence (AI) and the Internet of Things (IoT), modern surveillance systems have become more efficient and intelligent. AI-based computer vision techniques enable systems to analyze visual data, detect objects, and recognize faces in real time. IoT further enhances these systems by enabling communication between devices, remote monitoring, and instant alert generation [3].

Computer vision plays a crucial role in smart surveillance by enabling automatic face detection and recognition. It allows systems to identify individuals based on facial features without human intervention. This technology is particularly useful in access control systems where identifying authorized and unauthorized individuals is essential [4].

Recent developments in deep learning models, such as DeepFace and OpenCV-based detection techniques, have significantly improved the accuracy of face recognition systems. These models can recognize faces even under challenging conditions such as variations in lighting, facial expressions, and viewing angles. This makes them suitable for real-world applications in security systems [5], [6].

However, despite these advancements, many existing surveillance systems still rely on cloud-based processing. This introduces latency, increases dependency on internet connectivity, and raises concerns regarding data privacy and security. Continuous video storage also requires high bandwidth and large storage capacity, which increases system cost and complexity [7].

To overcome these challenges, edge computing has emerged as an effective solution. Edge computing allows data to be processed locally on devices, reducing latency and improving response time. It also enhances data privacy by minimizing the need to transmit sensitive information over external networks [8].

In addition to surveillance, modern systems require intelligent access control mechanisms. Storing images of authorized individuals in a structured database enables the system to maintain secure and efficient identification. This allows the system to recognize known individuals and differentiate them from unknown persons effectively [9].

Furthermore, integrating real-time notification mechanisms improves system reliability. When an unknown individual is detected, the system captures the image and sends instant alerts through email and Telegram notifications, enabling immediate action and response [10].

Therefore, this paper proposes an AI-enabled smart surveillance system that integrates face detection and recognition using OpenCV and DeepFace with edge computing. The system captures live video, identifies individuals, and generates alerts for unknown persons. The proposed system ensures enhanced security, real-time monitoring, improved accuracy, and reduced human intervention, making it suitable for modern smart home and security applications.

II. LITERATURE REVIEW

In recent years, intelligent surveillance systems have gained significant attention due to increasing security concerns and the limitations of traditional monitoring methods. Conventional surveillance systems rely on continuous video recording and manual observation, which are inefficient and often fail to provide real-time responses. These systems require constant human attention, which increases the chances of errors and missed events. As a result, researchers have focused on developing automated systems using computer vision techniques to improve accuracy, efficiency, and response time in surveillance applications [1].

Face detection is a fundamental component of modern surveillance systems. It is the first step in identifying individuals within an image or video stream. Various studies have implemented OpenCV-based approaches for real-time face detection due to their simplicity, flexibility, and effectiveness. Haar Cascade classifiers are widely used for detecting human faces as they provide fast detection speed and are

suitable for real-time applications. These methods are computationally efficient and can be easily integrated into live camera systems, making them ideal for practical implementations [2].

Face recognition technology has been significantly improved with the use of deep learning models. Approaches such as DeepFace utilize pre-trained neural networks to extract facial features and compare them with stored images. These models are capable of learning complex facial patterns and achieving high accuracy in identifying individuals. They perform well under varying conditions such as changes in lighting, facial expressions, partial occlusion, and different viewing angles. This makes deep learning-based recognition systems more reliable and suitable for real-world surveillance environments [3], [4].

The integration of the Internet of Things (IoT) in surveillance systems has enabled remote monitoring and real-time communication between devices. IoT-based systems allow users to access surveillance data from anywhere and receive instant alerts on their mobile devices. This improves system accessibility and enables users to take immediate action when suspicious activity is detected. The combination of IoT with AI-based systems has led to the development of smart surveillance solutions used in homes, offices, and public places [5].

Cloud storage plays an important role in managing and storing data in surveillance systems. It allows efficient storage of images and videos, including authorized user data such as family members' faces. Cloud-based solutions provide scalability, remote access, and easy data management. Users can update or modify stored data without affecting system performance. This makes cloud storage suitable for maintaining large databases in modern surveillance applications [6].

However, reliance on cloud-based systems introduces several challenges such as increased latency, dependency on internet connectivity, and potential data privacy risks. Continuous data transmission over networks can lead to delays in real-time processing and increase the chances of data breaches. To address these issues, edge computing has been introduced as an efficient alternative. In edge computing, data processing is performed locally on the device, reducing delay and improving system performance. It also enhances security by minimizing the need to transfer sensitive data over external networks [7].

Recent studies have also explored smart access control systems using face recognition technology. These systems allow access only to authorized individuals while restricting unknown users. Such systems are commonly used in smart homes and secure environments. Some advanced systems are designed to support remote approval mechanisms, where users can verify and control access through notifications. This approach improves both security and convenience, as users can manage access even when they are not physically present at the location [8].

Real-time alert systems using communication platforms such as email and messaging services (e.g., Telegram) have been widely adopted in surveillance applications. These systems provide immediate notifications when an unknown person is detected or suspicious activity occurs. Instant alerts enable users to respond quickly and take necessary actions, thereby improving the overall effectiveness and reliability of the surveillance system [9].

Despite these advancements, several challenges still exist in modern surveillance systems. These include maintaining accuracy under different environmental conditions, ensuring data privacy, reducing

system latency, and achieving efficient integration of multiple technologies. Therefore, there is a need for a comprehensive and integrated smart surveillance system that combines face detection, face recognition, edge computing, and real-time alert mechanisms to enhance security, efficiency, and reliability in modern applications.

III. METHODOLOGY

The proposed AI-enabled smart surveillance system is designed to perform real-time face detection and recognition using computer vision techniques. The system is implemented using Python, where OpenCV is used for face detection and DeepFace is used for face recognition. The system processes video input in real time and classifies individuals as known or unknown. It also stores captured images and generates alerts when required. The methodology is organized into multiple functional components for efficient system operation.

A. System Architecture

The system architecture consists of multiple modules that work together to perform real-time surveillance. The camera captures live video input, which is processed by the face detection and recognition modules. The system then compares detected faces with stored images and performs actions based on the result.

The system includes the following components:

- Camera module for capturing live video
- OpenCV for face detection
- DeepFace for face recognition
- Database for storing known faces
- Storage system for unknown faces
- Notification mechanism for alerts

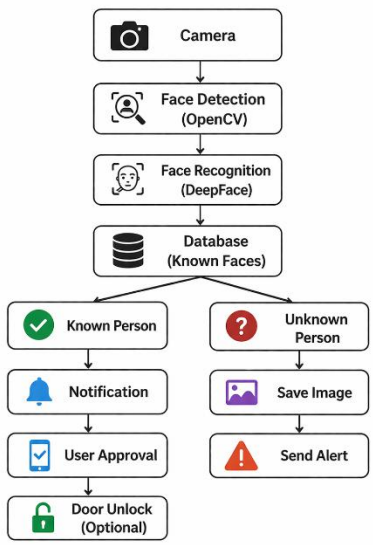

Fig. 1: System Architecture of Proposed Surveillance System

B. Face Detection and Recognition

The system uses OpenCV for detecting human faces in real-time video streams. Haar Cascade classifiers are applied to identify faces in each frame. Once a face is detected, the system extracts the region of interest (ROI) and sends it to the recognition module.

The recognition process is performed using the DeepFace framework. The detected face is compared with stored images of authorized users available in the database. Based on similarity matching, the system classifies the individual as either known or unknown. This step ensures accurate identification of individuals in the system.

C. Data Storage and Management

The system maintains a structured database of known individuals (family members) using stored images. These images are used for comparison during the recognition process. The data can be stored locally in folders for efficient access and management.

When an unknown individual is detected, the system captures the image and stores it in a separate folder. This allows the user to

review unknown entries later. The system ensures proper organization and handling of both known and unknown face data.

D. Notification and Alert Mechanism

The system includes a notification mechanism to inform users about detected individuals. When an unknown person is detected, the system generates an alert and can notify the user through communication platforms such as Email or Telegram.
This feature helps in improving security by allowing users to take immediate action when suspicious activity is detected. The alert system enhances the reliability and responsiveness of the surveillance system.

E. Edge-Based Processing

The system performs all processing tasks locally using edge computing techniques. This means that face detection and recognition are executed on the local device without relying heavily on cloud servers.
Edge-based processing reduces latency, improves system speed, and enhances data privacy. The system can also be deployed on devices such as Raspberry Pi for real-time edge-based surveillance applications.

F. Working Flow of the System

The working of the system can be summarized as follows:

1. The system captures live video using a camera.
2. Face detection is performed using OpenCV.
3. The detected face is extracted from the frame.
4. DeepFace compares the face with stored images.
5. The system identifies the person as known or unknown.
6. If the person is known, the system recognizes the individual.
7. If the person is unknown, the system captures and stores the image.
8. An alert can be generated to notify the user.
9. The system continues monitoring in real time.

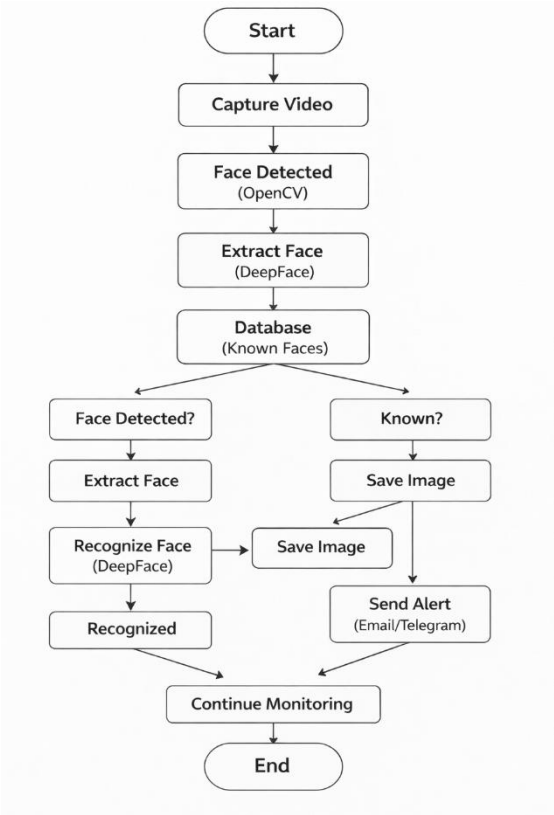

Fig. 2: Flowchart of Proposed System

G. Future Enhancement

The system is designed to support additional advanced features such as cloud-based data storage and remote approval-based access control. In this approach, authorized user images can be stored in cloud storage, and users can approve or reject access requests through notifications. Additionally, a smart door lock mechanism can be integrated, where access can be granted based on user approval. These features can enhance system functionality and provide a more secure and flexible surveillance solution.

IV. RESULTS

The proposed smart surveillance system was successfully implemented and tested for real-time face detection and recognition.

The system demonstrates effective performance in identifying individuals, storing unknown faces, and generating alerts when required. The experimental results validate the efficiency and reliability of the system.

A. Functional Performance

The system was evaluated based on its ability to detect faces in real-time video streams. The OpenCV-based detection module performs efficiently and accurately under normal conditions.

- The system successfully detects faces from live camera input.
- Real-time processing ensures continuous monitoring.
- The detection speed is fast with minimal delay.

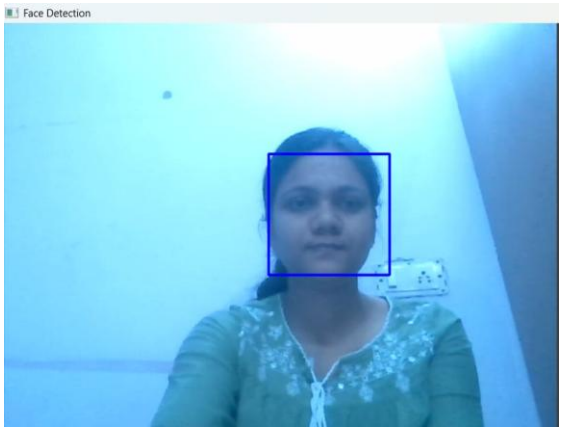

Fig. 3: Face Detection Output

B. Face Recognition Accuracy

The DeepFace model was tested for recognizing known individuals using stored images. The system provides accurate results in identifying authorized users.

- Known individuals are correctly recognized.
- Recognition works under slight variations in lighting and facial expressions.
- Minimal misclassification was observed during testing.

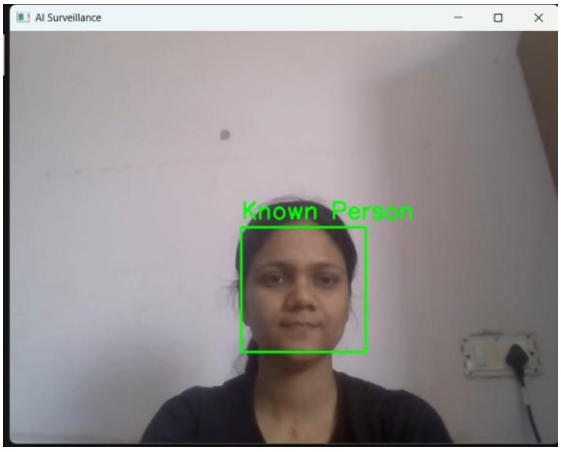

Fig. 4: Known Face Recognition

C. Unknown Person Detection and Storage

The system effectively detects and handles unknown individuals by capturing and storing their images.

- Unknown faces are automatically captured.
- Images are stored in a separate folder for future reference.
- This feature enhances system security.

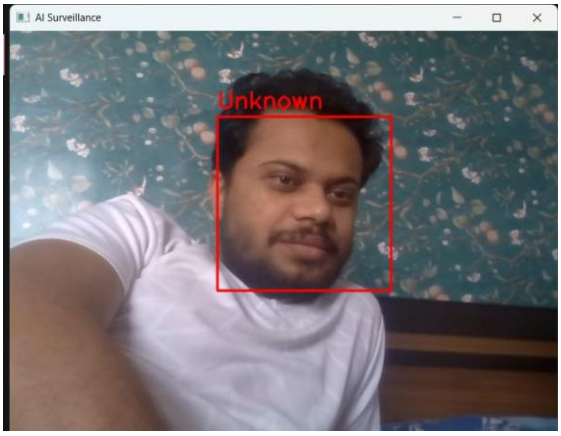

Fig. 5: Unknown Face Detection

D. Alert and Notification Performance

The system supports alert generation when an unknown person is detected.

- Alerts can be generated to notify the user.
- Notifications can be sent via Email.

- This allows quick response to suspicious activity.

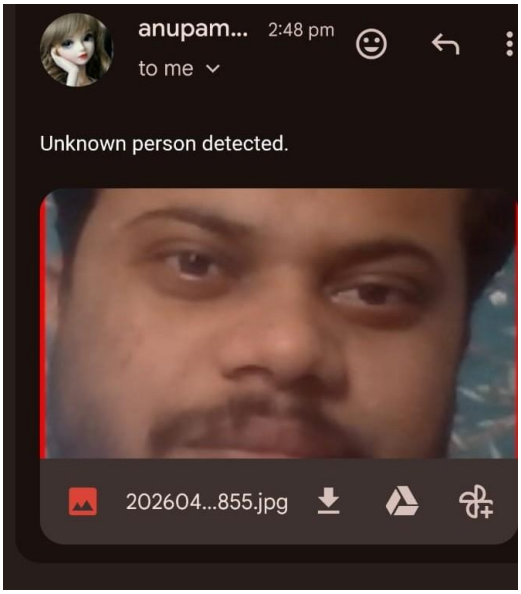

Fig. 6: Alert Notification Email

E. System Efficiency

The overall system performance was evaluated in terms of speed and reliability.

- The system operates continuously without interruption.
- Real-time processing ensures quick response.
- Local processing reduces latency and improves efficiency.

The results confirm that the proposed system is efficient, reliable, and suitable for real-time smart surveillance applications.

V. CONCLUSION

The proposed AI-enabled smart surveillance system successfully demonstrates the application of computer vision techniques for real-time face detection and recognition. By integrating OpenCV for face detection and DeepFace for face recognition, the system is able to accurately identify known individuals and distinguish unknown persons, thereby enhancing security and monitoring capabilities. The experimental results indicate that the system performs efficiently with minimal processing delay while handling live video streams. The use of automated detection reduces the need for continuous human monitoring and minimizes the chances of missing important events, thereby improving the reliability and effectiveness of the surveillance system in practical environments. In addition, the system is capable of capturing and storing images of unknown individuals, which can be used for future reference and security analysis. The alert generation mechanism further enhances system functionality by enabling users to receive notifications in case of suspicious activity, allowing timely response and decision-making. The implementation of edge-based processing ensures faster performance and reduces dependency on cloud-based systems while also improving data privacy by processing sensitive information locally without transmitting it over external networks. This makes the system suitable for real-time and secure applications in smart homes and small-scale environments. Although the current system focuses on face detection and recognition, it provides a flexible framework for future enhancements. Overall, the proposed system offers a cost-effective, efficient, and intelligent solution for modern surveillance applications with the potential to be extended for advanced security features.

VI. FUTURE WORK

The proposed system can be further enhanced by integrating advanced features to improve its functionality and usability. One of the major improvements includes the implementation of a smart door lock system, where access can be granted or denied based on user approval, thereby enabling automated access control and enhancing security in real-world applications. Additionally, the system can be integrated with cloud storage to improve data management and enable remote accessibility, allowing users to access stored data from anywhere and maintain a

centralized database of authorized individuals. The system can also be deployed on embedded devices such as Raspberry Pi to make it more compact, cost-effective, and suitable for real-time smart home environments. Furthermore, mobile application integration can be introduced to provide better user interaction, remote monitoring, and control over the system. Future enhancements may also include multi-face detection, improved recognition accuracy, and advanced security features to make the system more robust, scalable, and efficient.

VII. REFERENCES

[1] P. Viola and M. Jones, "Rapid Object Detection using a Boosted Cascade of Simple Features," *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2001.

[2] R. Lienhart and J. Maydt, "An Extended Set of Haar-like Features for Rapid Object Detection," *Proc. IEEE Int. Conf. Image Process.*, 2002.

[3] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

[4] O. M. Parkhi, A. Vedaldi, and A. Zisserman, "Deep Face Recognition," *Proc. British Mach. Vis. Conf.*, 2015.

[5] S. Taigman, M. Yang, M. Ranzato, and L. Wolf, "DeepFace: Closing the Gap to Human-Level Performance in Face Verification," *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2014.

[6] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," *Adv. Neural Inf. Process. Syst.*, 2012.

[7] W. Shi and S. Dustdar, "The Promise of Edge Computing," *Computer*, vol. 49, no. 5, pp. 78–81, 2016.

[8] L. Atzori, A. Iera, and G. Morabito, "The Internet of Things: A Survey," *Comput. Netw.*, vol. 54, no. 15, pp. 2787–2805,2010.

[9] D. Zhang, L. Wang, and G. Pan, "A Survey on Human Activity Recognition using Wearable Sensors," *IEEE Commun. Surveys Tuts.*, 2018.

[10] S. S. S. Krishnan, "Smart Surveillance System using IoT and Computer Vision," *Int. J. Eng. Res. Technol.*, 2020.